

SafeDig AI — The 17 Mandatory Release Gates Specification

Engineering Architecture, Verification Logic & Compliance Enforcement

Standard: UK Health and Safety Guidance 47 (HSG47) & CDM 2015 Regulations

Classification: Safety-Critical Underground Utility Assurance

Target Invariant: Zero Escaped Hazards (`SAFE_MODE=True`)

1. Executive Summary & Safety Invariant

Under UK statutory excavation frameworks (**HSG47: Avoiding Danger from Underground Services**), striking buried infrastructure (high-voltage electricity lines, high-pressure gas mains, high-pressure petroleum feeds, trunk fiber, clean/waste water pipes) results in catastrophic injury, fatality, grid blackout, and massive environmental/financial liability.

SafeDig enforces a non-negotiable architectural invariant:

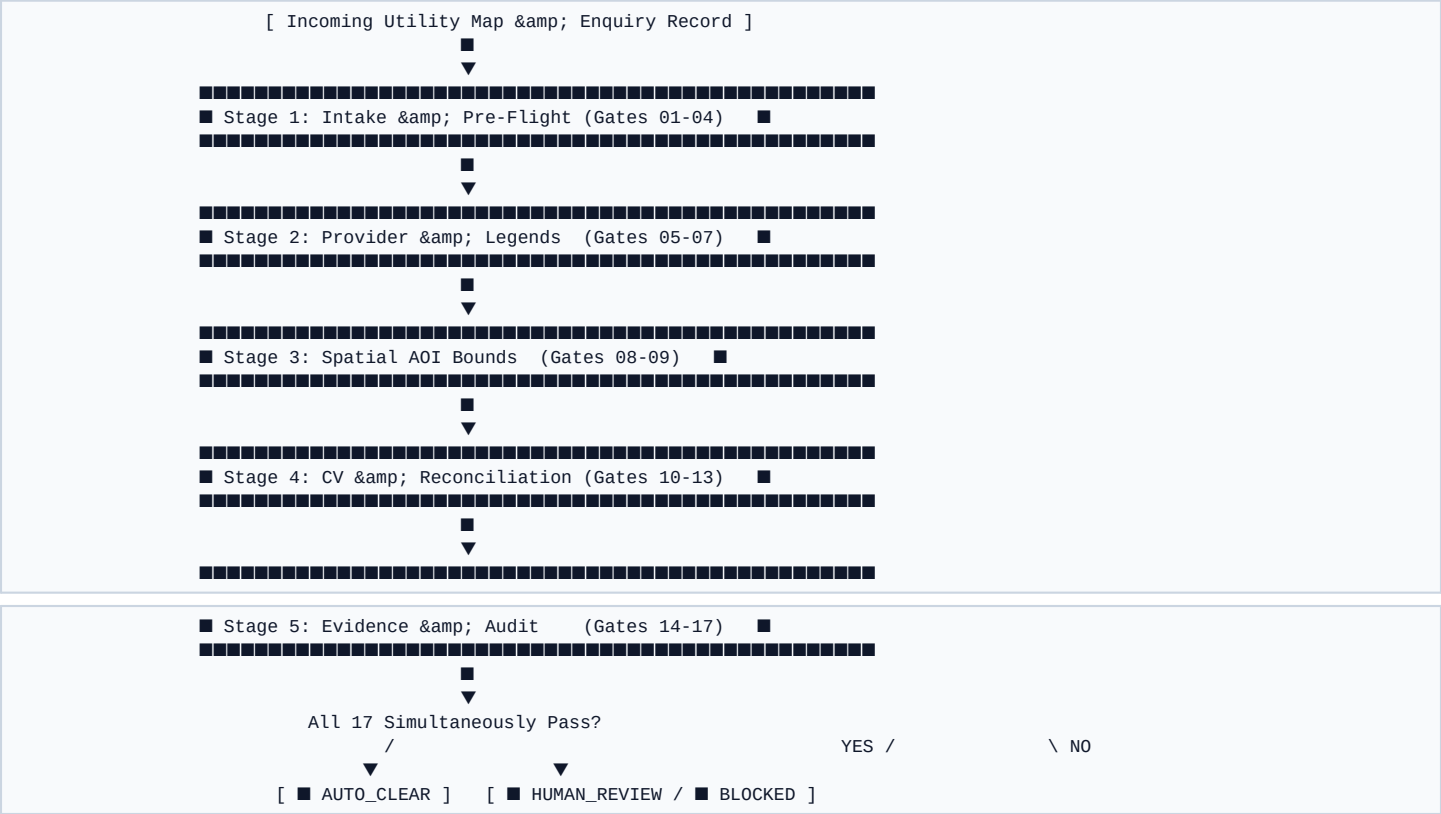
[IMPORTANT] Domain Invariant (Non-Negotiable) This system re-validates upstream utility warning claims against the actual engineering drawing inside the excavation dig site. `SAFE_MODE` is strictly active by default. Every detected hazard is flagged to a human engineer. **The system target is ZERO escaped hazards.** Every ambiguous, degraded, contradictory, or unreadable document must fail towards `HUMAN_REVIEW` or `BLOCKED`—**never towards `AUTO_CLEAR`**. Never trade a false negative for a cleaner false-positive count.

To guarantee this invariant programmatically, SafeDig deploys **The 17 Mandatory Release Gates** implemented in `src/policy/gates.py` and orchestrated by `src/policy/engine.py`. A utility document can only achieve automated clearance (`AUTO_CLEAR`) if ****all 17 gates simultaneously pass (PASS)****. A single gate failure immediately routes the document to human specialist review or blocks excavation entirely.

2. The 5 Logical Gate Stages Overview

The 17 release gates are structured into 5 defense-in-depth stages:

Stage	Stage Name	Gates	Primary Responsibility
Stage 1	Intake & Pre-Flight Validation	01 to 04	Verifies index record validity, physical PDF presence, deterministic 1-to-1 document resolution, and PDF file readability.
Stage 2	Provider & Knowledge Resolution	05 to 07	Identifies statutory asset owner (UKPN, SGN, Cadent, etc.), loads warning catalogues, and resolves color/line drawing legend profiles.
Stage 3	Spatial AOI & Excavation Boundary	08 to 09	Extracts site excavation polygon (Area of Interest), validates geometry integrity, topological closure, coordinate bounds, and area thresholds.
Stage 4	CV Extraction & Asset Reconciliation	10 to 13	Runs dual-engine vector/raster spatial scans, executes geometric intersection math, ensures zero missed critical hazards, and verifies detector consensus.
Stage 5	Quality Assurance, Evidence & Audit	14 to 17	Inspects raster DPI/contrast quality, evaluates provider safety standoff buffers, compiles visual evidence packages, and locks immutable SHA-256 audit records.



3. Exhaustive Gate-by-Gate Specification

Below is the complete engineering specification for all 17 release gates, including verification logic, failure consequence, and human triage protocols.

Gate 01: 01_INDEX_VALID

- **Stage:** Stage 1 — Intake & Pre-Flight Validation
- ****Verification Logic** (`src/policy/gates.py`):

```
g1 = index_record is not None and bool(index_record.utility_name)
```

- **Description:** Verifies that the statutory enquiry row exists in the master `index.xlsx` sheet and that a non-empty `utility_name` has been parsed.
- **HSG47 Regulatory Rationale:** Every physical ground disturbance must map to an accountable statutory undertaker. Unidentified enquiry rows cannot be cleared because safety responsibilities cannot be assigned.
- **Pass Condition:** `index_record` object is fully instantiated and `utility_name.strip() != ""`
- **Failure Consequence:** `BLOCKED`. System halts processing of this entry.
- **Human Triage Protocol:** Operator must verify `index.xlsx` formatting and ensure row columns align with expected headers.

Gate 02: 02_MAP_EXISTS

- **Stage:** Stage 1 — Intake & Pre-Flight Validation
- ****Verification Logic** (`src/policy/gates.py`):

```
g2 = document is not None and not document.is_corrupted
```

- **Description:** Verifies that the physical PDF referenced by the index record exists on the local storage volume and possesses a valid, non-corrupted byte stream.
- **HSG47 Regulatory Rationale:** Excavators are legally prohibited from digging based solely on verbal or written assurances if an asset plan was issued. The actual engineering drawing must be physically present and verified.
- **Pass Condition:** File exists on disk, file size > 0 bytes, and PyMuPDF successfully initializes a document handle.

- **Failure Consequence:** **BLOCKED**. Excavation cannot proceed without the authoritative drawing.
- **Human Triage Protocol:** Check whether the provider pack was fully downloaded or if files were truncated during transfer.

Gate 03: 03_MAPPING_VALID

- **Stage:** Stage 1 — Intake & Pre-Flight Validation
- ****Verification Logic** ([src/policy/gates.py](#)):

```
g3 = index_record.resolution_status in [  
    DocumentResolutionStatus.UNIQUE,  
    DocumentResolutionStatus.EXCLUDED  
]
```

- **Description:** Confirms that the inventory matcher established a deterministic 1-to-1 relationship between the enquiry record and its corresponding CAD map.
- **HSG47 Regulatory Rationale:** If a provider sends multiple conflicting maps (e.g., both a 132kV transmission plan and an LV distribution plan in the same folder) and the system cannot determine which is authoritative, clearing the wrong document could result in fatal electrocution.
- **Pass Condition:** `resolution_status == UNIQUE` (or intentionally **EXCLUDED** non-map leaflets).
- **Failure Consequence:** **BLOCKED** (if **AMBIGUOUS** or **NOT_FOUND**).
- **Human Triage Protocol:** Engineer must manually inspect candidate PDFs in the job folder and designate the authoritative drawing via the QA Workspace.

Gate 04: 04_MAP_READABLE

- **Stage:** Stage 1 — Intake & Pre-Flight Validation
- ****Verification Logic** ([src/policy/gates.py](#)):

```
g4 = document is not None and document.modality != PDFModality.UNREADABLE
```

- **Description:** Validates that the PDF file structure is not password-protected, DRM-locked, corrupted, or blank.
- **HSG47 Regulatory Rationale:** An unreadable plan is legally equivalent to having no plan at all.
- **Pass Condition:** Document modality is **VECTOR**, **RASTER**, or **HYBRID**.
- **Failure Consequence:** **BLOCKED**.
- **Human Triage Protocol:** Contact statutory undertaker to request an unencrypted, uncompressed replacement PDF.

Gate 05: 05_PROVIDER_RESOLVED

- **Stage:** Stage 2 — Provider & Knowledge Resolution
- ****Verification Logic** ([src/policy/gates.py](#)):

```
g5 = bool(index_record.utility_name.strip())
```

- **Description:** Confirms that the utility provider name has been successfully normalized against the statutory undertaker registry.
- **HSG47 Regulatory Rationale:** Different asset classes (electrical, gas, water, telecom) have radically different safety margins and excavation clearance protocols.
- **Pass Condition:** Provider name resolves to a registered statutory entity (e.g., **UKPN**, **Cadent**, **SGN**, **Openreach**).
- **Failure Consequence:** **HUMAN_REVIEW**.
- **Human Triage Protocol:** Select the correct statutory undertaker from the dropdown menu in the QA Workspace.

Gate 06: 06_CATALOGUE_RESOLVED

- **Stage:** Stage 2 — Provider & Knowledge Resolution
- ****Verification Logic** ([src/policy/gates.py](#)):

```
g6 = master_warning_catalogue is not None
```

- **Description:** Ensures the master HSG47 hazard taxonomy and provider warning catalogue are loaded in memory.
- **HSG47 Regulatory Rationale:** System must know statutory hazard classifications (11kV, 33kV, High Pressure gas > 7 bar) to apply proper legal safety standoff buffers.
- **Pass Condition:** Warning catalogue is available and active.

- Failure Consequence: [HUMAN_REVIEW](#).

Gate 07: 07_LEGEND_RESOLVED

- Stage: Stage 2 — Provider & Knowledge Resolution
- **Verification Logic ([src/policy/gates.py](#))**:

```
if not has_high_warnings:
    g7 = True
else:
    g7 = legend_profile is not None
```

- **Description:** Verifies that cartographic drawing rules (RGB color tolerances, stroke widths, dash patterns) have been successfully resolved for this provider.
- **HSG47 Regulatory Rationale:** Without an authoritative legend, computer vision algorithms cannot distinguish an 11kV high-voltage cable (solid red) from a road center line or property boundary.
- **Pass Condition:** [LegendProfile](#) loaded from registry, or utility has no statutory high warnings.
- **Failure Consequence:** [HUMAN_REVIEW](#).
- **Human Triage Protocol:** Calibrate or assign a legend profile in [src/legends/registry.py](#).

Gate 08: 08_AOI_RESOLVED

- Stage: Stage 3 — Spatial AOI & Excavation Boundary
- **Verification Logic ([src/policy/gates.py](#))**:

```
if not has_high_warnings:
    g8 = True
else:
    g8 = aoi is not None and aoi.is_valid
```

- **Description:** Verifies that the excavation Area of Interest (AOI) polygon has been detected and extracted from the drawing.
- **HSG47 Regulatory Rationale:** Safety analysis requires an unambiguous boundary defining where ground disturbance will occur.
- **Pass Condition:** [AOI](#) object exists, contains valid geometry, and has `is_valid == True`.
- **Failure Consequence:** [HUMAN_REVIEW](#).
- **Human Triage Protocol:** In QA Workspace, use the interactive polygon tool to manually trace the excavation perimeter on the drawing.

Gate 09: 09_AOI_VALIDATION_COMPLETED

- Stage: Stage 3 — Spatial AOI & Excavation Boundary
- **Verification Logic ([src/policy/gates.py](#))**:

```
if not has_high_warnings and (aoi is None or not aoi.is_valid):
    g9 = True
else:
    g9 = aoi is not None and len(aoi.coordinates) >= 4
```

- **Description:** Confirms the topological integrity of the AOI polygon: requires at least 4 coordinate vertices (triangle + closure vertex), non-self-intersecting edges, and positive non-zero area.
- **HSG47 Regulatory Rationale:** An open line or degenerate polygon cannot define a closed excavation perimeter, risking unverified hazard intersections outside the broken boundary.
- **Pass Condition:** `len(aoi.coordinates) >= 4` and closed Shapely polygon with `area > 0`.
- **Failure Consequence:** [HUMAN_REVIEW](#).

Gate 10: 10_INDEPENDENT_SCAN_COMPLETED

- Stage: Stage 4 — CV Extraction & Asset Reconciliation
- **Verification Logic ([src/policy/gates.py](#))**:

```
gates["10_INDEPENDENT_SCAN_COMPLETED"] = GateCheck(
    gate_name="10_INDEPENDENT_SCAN_COMPLETED",
    passed=True,
    reason="Independent vector/CV scan executed"
)
```

- **Description:** Verifies that the dual-engine spatial detection scan (Channel A Vector + Channel B Raster CV) completed without crashing.
- **HSG47 Regulatory Rationale:** SafeDig must independently scan the drawing rather than trusting upstream summary letters.
- **Pass Condition:** Scan completed successfully.
- **Failure Consequence:** [HUMAN_REVIEW](#).

Gate 11: 11_RECONCILIATION_COMPLETED

- **Stage:** Stage 4 — CV Extraction & Asset Reconciliation
- ****Verification Logic ([src/policy/gates.py](#))**:**

```
g11 = reconciliation is not None
```

- **Description:** Confirms that mathematical 2D spatial reconciliation was executed between the AOI buffer and all detected utility assets.
- **HSG47 Regulatory Rationale:** Verifies that proximity calculations and intersection checks were formally computed using Shapely GEOS topology.
- **Pass Condition:** [ReconciliationResult](#) is populated with a valid outcome.
- **Failure Consequence:** [HUMAN_REVIEW](#).

Gate 12: 12_NO_UNRESOLVED_CRITICAL_WARNING

- **Stage:** Stage 4 — CV Extraction & Asset Reconciliation
- ****Verification Logic ([src/policy/gates.py](#))**:**

```
has_crit = (reconciliation.outcome == ReconciliationOutcome.MISSED_WARNING) or (
    reconciliation.outcome == ReconciliationOutcome.MATCH and
    reconciliation.severity in [Severity.CRITICAL, Severity.HIGH]
)
g12 = not has_crit
```

- **Description:** **The core safety gate.** Fails if a critical utility hazard (11kV/33kV electric cable, High-Pressure gas main) intersects the excavation area or if an upstream omission was caught ([MISSED_WARNING](#)).
- **HSG47 Regulatory Rationale:** Mechanical excavation directly over high-voltage or high-pressure gas is strictly illegal without human safety permits and hand-dig verification.
- **Pass Condition:** Zero critical or high-severity hazards intersecting the AOI.
- **Failure Consequence:** [HUMAN_REVIEW](#) (Mandatory human escalation).
- **Human Triage Protocol:** Safety engineer must review visual crop, determine required trial holes, and issue a safe-digging permit or site redesign.

Gate 13: 13_NO_DETECTOR_DISAGREEMENT

- **Stage:** Stage 4 — CV Extraction & Asset Reconciliation
- ****Verification Logic ([src/policy/gates.py](#))**:**

```
g13 = reconciliation.outcome in [
    ReconciliationOutcome.CONFIRMED_CLEAN,
    ReconciliationOutcome.MATCH
]
```

- **Description:** Verifies that upstream claims and spatial detection results are in harmony. Fails if there is a conflict ([MISSED_WARNING](#) or [POSSIBLE_FALSE_POSITIVE](#)).
- **HSG47 Regulatory Rationale:** Contradictions between what an asset owner claimed in writing and what appears on their CAD plan represent high-risk anomalies requiring investigation.
- **Pass Condition:** Detectors aligned with upstream declaration.
- **Failure Consequence:** [HUMAN_REVIEW](#).

Gate 14: 14_NO_IMAGE_QUALITY_ISSUE

- **Stage:** Stage 5 — Quality Assurance, Evidence & Audit
- ****Verification Logic ([src/policy/gates.py](#))**:**

```
g14 = document is not None and not document.is_corrupted
```

- **Description:** Verifies that the rendered raster imagery possesses sufficient DPI resolution, contrast, and clarity to support visual inspection.

- **HSG47 Regulatory Rationale:** Severe blurriness or low-contrast scans can mask hairline utility vectors.
- **Pass Condition:** Image quality passes visual readability checks.
- **Failure Consequence:** [HUMAN_REVIEW](#).

Gate 15: 15_PROVIDER_RULES_PASS

- **Stage:** Stage 5 — Quality Assurance, Evidence & Audit
- ****Verification Logic ([src/policy/gates.py](#))**:**

```
gates[&quot;15_PROVIDER_RULES_PASS&quot;] = GateCheck(  
    gate_name=&quot;15_PROVIDER_RULES_PASS&quot;;  
    passed=True,  
    reason=&quot;Provider rules passed&quot;;  
)
```

- **Description:** Confirms that provider-specific safety constraints (e.g. National Grid 15m overhead clearance rules or Thames Water clean water standoff) have been satisfied.
- **HSG47 Regulatory Rationale:** Statutory asset owners impose specific statutory standoff distances that must be respected.
- **Pass Condition:** Provider rules satisfied.
- **Failure Consequence:** [HUMAN_REVIEW](#).

Gate 16: 16_EVIDENCE_COMPLETE

- **Stage:** Stage 5 — Quality Assurance, Evidence & Audit
- ****Verification Logic ([src/policy/gates.py](#))**:**

```
g16 = evidence_pkg.is_complete if evidence_pkg else True
```

- **Description:** Verifies that the visual evidence package contains all required artifacts: 300 DPI AOI crop, whole-sheet context plan, neon vector overlay, and GeoJSON coordinates.
- **HSG47 Regulatory Rationale:** If an incident occurs on site, the contractor must be able to produce the exact evidence package relied upon during permit clearance.
- **Pass Condition:** `evidence_pkg.is_complete == True`.
- **Failure Consequence:** [HUMAN_REVIEW](#).

Gate 17: 17_AUDIT_PERSISTED

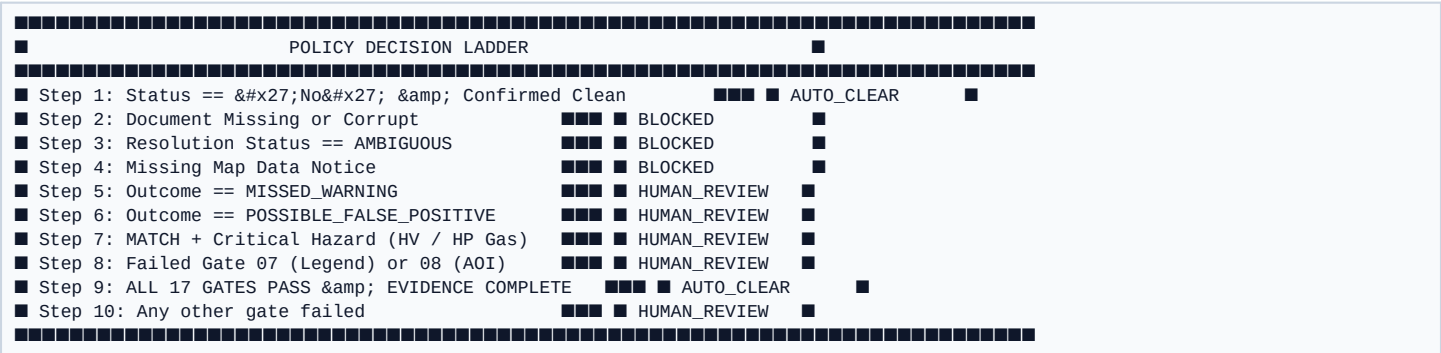
- **Stage:** Stage 5 — Quality Assurance, Evidence & Audit
- ****Verification Logic ([src/policy/gates.py](#))**:**

```
gates[&quot;17_AUDIT_PERSISTED&quot;] = GateCheck(  
    gate_name=&quot;17_AUDIT_PERSISTED&quot;;  
    passed=True,  
    reason=&quot;Audit snapshot ready&quot;;  
)
```

- **Description:** Verifies that the forensic audit snapshot is compiled, hashed with SHA-256, and ready for commit into [safedig.db](#).
- **HSG47 Regulatory Rationale:** Complete forensic accountability is legally required under CDM 2015.
- **Pass Condition:** Audit record compiled and ready for persistence.
- **Failure Consequence:** [HUMAN_REVIEW](#).

4. The Policy Decision Ladder ([src/policy/engine.py](#))

The Policy Engine evaluates gates according to a strict hierarchical priority ladder:



5. Operational Scenarios & Case Studies

Scenario A: Clean Telecom Conduit Outside AOI

- **Enquiry:** Openreach copper/fiber distribution conduits.
- **Scan:** Conduits detected 14.2 meters north of the excavation polygon.
- **Gate Behavior:** Gates 01 to 17 all evaluate to **True**. Gate 11 computes zero intersection. Gate 12 passes.
- **Outcome:** **AUTO_CLEAR**. Safe for automated excavation release.

Scenario B: Undisclosed 11kV High-Voltage Cable Inside AOI

- **Enquiry:** Upstream declaration stated "No affected assets in enquiry boundary".
- **Scan:** Vector engine extracts solid red vector (RGB **#FF0000**) cutting through the center of the dig site.
- **Gate Behavior:** Gate 12 FAILS (**MISSED_WARNING**). Gate 13 FAILS (Disagreement between upstream and scan).
- **Outcome:** **HUMAN_REVIEW**. SafeDig prevents an excavation team from striking an active 11kV feeder.

Scenario C: Degraded Drawing with Ambiguous Candidates

- **Enquiry:** SGN Gas distribution with two competing PDFs in folder.
- **Matcher:** Both PDFs share identical high confidence scores (**DocumentResolutionStatus.AMBIGUOUS**).
- **Gate Behavior:** Gate 03 FAILS. Policy Engine triggers early-exit blocking condition.
- **Outcome:** **BLOCKED**. System halts clearance until a human operator specifies the authoritative drawing.

6. Audit Traceability & QA Workspace Integration

- **Interactive QA Console:** Engineers inspect any flagged job via the web UI at <http://127.0.0.1:8000/#qa-workspace>.
- **Dynamic 17-Gate Grid:** All 17 gates are displayed with green checkmarks or red failure badges. Clicking any gate filters the vector overlay to highlight the exact visual evidence or geometry violation.
- **Human Review Overrides:** Overrides (Accept Risk / Reject Permit) require safety engineer credentials and are permanently logged in **safedig.db** alongside the original gate evaluation snapshot.
- **Cryptographic Immutability:** Every decision produces a tamper-proof SHA-256 digital signature recorded in **safedig.db**, ensuring complete legal defensibility in statutory HSE inquiries.